

Anthony Ibarra | Electrical & Computer Engineer

(773)387-8485 : 28aibarra@gmail.com : [github/Portfolio](https://github.com/28aibarra) : [Linkedin/anthony-i](https://www.linkedin.com/in/anthony-ibarra/)

Summary

Electrical and Computer Engineering graduate looking for the opportunity to become a professional in my respective career. Showcasing a strong focus in Embedded Systems, Control theory, Robotics, including pattern recognition involving AI CNN machine learning. And power electronics making me a flexible Embedded Software Engineer with a strong foundation. I possess various knowledge in different fields providing fast adaptable capability within the work environment of a multidisciplinary team. With the eagerness to learn on and off the job, demonstrating reliability and trustworthiness to depend on one another.

EDUCATION

University of Illinois Chicago (UIC)

Jan 2023 - May 2025

Master of Science in Electrical and Computer Engineering

Relevant Coursework: Audio & Acoustic Signal Processing, Adaptive Digital Filters, Linear Systems Theory & Design, Advanced Computer Communication, Convex Optimization, Intro Neural Networks

University of Illinois Chicago (UIC)

Aug 2017 - May 2022

Bachelor of Science in Computer Engineering

Relevant Coursework: Artificial Intelligence I, Principles of Modern Control & Principles of Auto Control, Pattern Recognition I, Computer Comm Networks I, Embedded Systems

SKILLS

Computer programming: Python, C/C++/C# , Ubuntu, AI (artificial intelligence) & ML(machine learning)

Software Knowledge: VS Code, GitHub, MATLAB, Quartus, code-composer, Arduino IDE, VNC viewer(raspberry Pi), (Microsoft Word, PowerPoint, Excel, etc.)

Speech: B1 Proficient in spanish

Academic Projects

Self Driving Car, Mechatronics Embedded Design

Jan 2023 - May 2023

- Lead the team and manage all time constraint tasks, development and design tasks, make timely decisions for the team's success in completing the product and hardware development
- Design a motor driver circuit through FET Driver implementation, either of (single fet, half bridge, or full H-bridge), controlled via a PWM input signal from the microcontroller
- Design a Boost Converter for the implementation multiple power systems
- Product testing using Oscilloscope, multimeters, benchtop powersupplies; and Soldering fixes when necessary
- Develop and tune controllers for both the steering and velocity controls (Servo motor, DC motor) respectively
- Design, develop the PCB through Altium Designer and get it manufactured, providing all necessary files such as BOM, gerber files, etc.
- Solder all through hole and SMT components including masking for PCB and Prototype; create any needed jumper wires for connections through crimping and soldering
- Prototype implementation of a perf board circuit, also used as an emergency backup board for PCB failure ensuring continuation of product development
- Implement Sensors and Encoders a filter for the line Camera or velocity measured input through embedded software development and test through debugging microcontroller software

Wildfire Classification, Intro to Neural Networks

Jan 2023 - May 2023

- Design a CNN (convolutional neural network) for pattern recognition and classification of real life cases
- Implement classification training for wildfires as it is similarly done for facial recognition
- Configure and tune the training and testing for the cnn model using python and pytorch with cuda cores and GPU's when able
- Use the created image classification model for wildfire prevention in testing

Automated Watering System, Senior Design

August 2021 - May 2022

- Work in a four-student team to prototype a product implementing a automated watering systems plants by taking moisture levels, outdoor weather conditions, and plant information data into account
- Manage team through verbal and written communication to make sure all assignments and goals are completed on time, and submit weekly reports based on progress of project development
- Embedded programming for a Arduino Nano iot 33 to consider watering decisions per plants moisture levels and water consumption
- Create mobile APP (Kivy framework) to display necessary information regarding the system, plants moisture levels and weather conditions
- Create a UDP client-server connection for communication between powered product and the mobile application; providing manual control over a wireless connection
- Implement wireless communication through Bluetooth Protocol (BLE) for consumer and their device